

**Tutorial-3**  
**ME-101, (Engineering Mechanics, Semester-II)**  
**Date: 5<sup>th</sup> Feb 2024 and Time: 7.55 to 8.50 AM**

**Name:** \_\_\_\_\_ **Roll No.** \_\_\_\_\_

**Important Note:** Write your final answers *ONLY* in the relevant blank spaces provided. All the associated calculations and rough work (if any), are to be shown in detail in a separate sheet and the same has to be attached with the question paper during submission. All numerical answers must accompany appropriate units.

**Q1.** Determine the moment  $M_A$  which must be applied at **A** to keep the frame in static equilibrium in the position, shown in Figure 1. Also calculate the magnitude of the pin reaction at **A**.

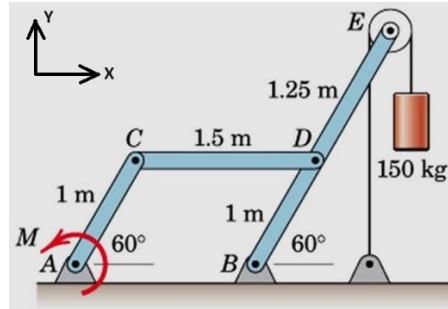


Figure 1

Solution of Q1:

Step 1: Draw the Free Body Diagram for the pulley **E** below.

[1 Mark]

Step 2: Draw the Free Body Diagram of the members, **AC**, **CD**, and **BE** separately below.

[3 Marks]

Step 3: Force in the member **CD** is \_\_\_\_\_. (assume Tensile as positive)

[2 Marks]

Step 4: The moment  $M_A$  = \_\_\_\_\_.

[2 Marks]

Step 5: Reactions at Pin **A**:  $R_{Ax}$  = \_\_\_\_\_ and  $R_{Ay}$  = \_\_\_\_\_.

[2 Marks]

(Refer X-Y axis mentioned in Figure-1 to decide the signs of the forces and moments)

Q2. Replace the distributed loading on the beam shown in Figure 2 with an equivalent resultant force, and specify its location on the beam measured from point A.

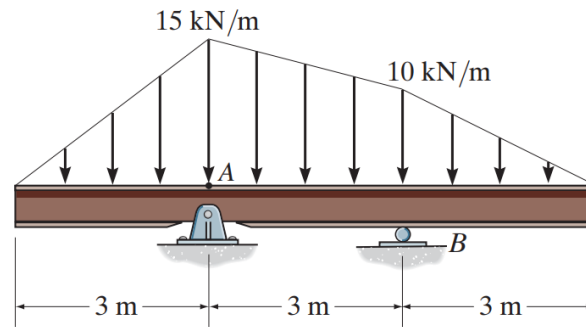


Figure 2

Solution of Q2:

Step 1: Draw the Free Body Diagram of the entire beam below.

[1 Mark]

Step 2: Reactions at A:  $R_{Ax} =$  \_\_\_\_\_ and  $R_{Ay} =$  \_\_\_\_\_.

[2 Marks]

Step 3: Reactions at B:  $R_{By} =$  \_\_\_\_\_.

[2 Marks]

Step 4: Equivalent Resultant Force (magnitude & sense) = \_\_\_\_\_

\_\_\_\_\_.

[2.5 Marks]

Step 4: Distance of line of action of the equivalent resultant force from A = \_\_\_\_\_

\_\_\_\_\_.

[2.5 Marks]

Q3. For the shaded area as shown in Figure 3, calculate the following:

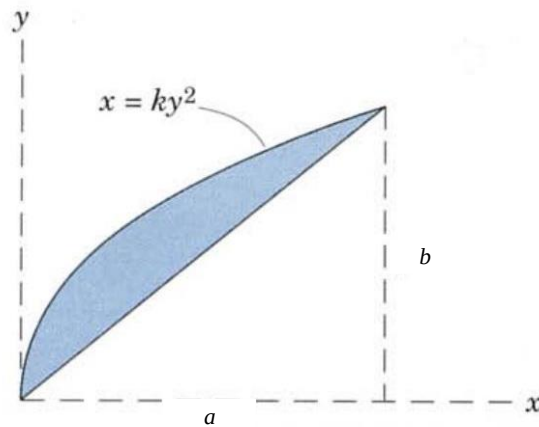


Figure 3

Solution of Q3:

- The area of the shaded portion shown in Fig. 3, in terms of  $a$  and  $b$  = \_\_\_\_\_ [3 Marks]
- The  $x$ -coordinate of the centroid of the shaded area in terms of  $a$  and  $b$  = \_\_\_\_\_ [3 Marks]
- The  $y$ -coordinate of the centroid of the shaded area in terms of  $a$  and  $b$  = \_\_\_\_\_ [3 Marks]
- The co-ordinates of the centroid, for  $a = 50$  mm and  $b = 40$  mm is \_\_\_\_\_ [1 Mark]

Q4. Figure 4 shows a concrete dam, whose factor of safety for tipping is defined as the ratio of the stabilizing moment due to the dam's weight divided by the overturning moment about  $O$  due to the water pressure.

- Draw the Free Body Diagram of the given figure [2 Mark]
- Calculate the pressure at the base of the dam and intensities of the load at the base of the dam [2 Mark]
- Calculate the dam's weight and the point of application [2 Marks]
- Calculate the force due to water pressure and the point of application [2 Marks]
- Calculate the overturning moment and stabilizing moment [2 Marks]

The concrete has a density  $\rho_c = 2.5 \text{ Mg/m}^3$  and for water the density is  $\rho_w = 1 \text{ Mg/m}^3$ .

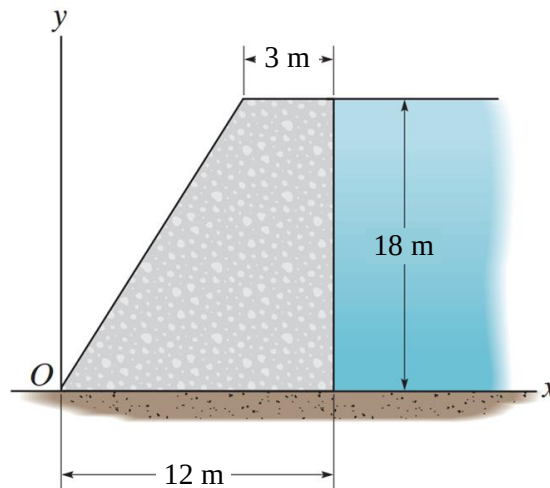


Figure 4

Solution of Q4:

- Step 1: Draw the Free Body Diagram of the concrete dam below. [2 Mark]

- The pressure at the base of the dam is \_\_\_\_\_ and the load intensity at the base of the dam is \_\_\_\_\_. [2 Mark]

- Step 3: The weight of the dam is \_\_\_\_\_ and the point of application of the weight is \_\_\_\_\_. [2 Marks]

- Step 4: The force on the dam due to water pressure is \_\_\_\_\_ and its point of application is \_\_\_\_\_. [2 Marks]

- The overturning moment is \_\_\_\_\_ and the stabilizing moment is \_\_\_\_\_. [2 Marks]